Respiration

15a

The respiration process – an explanation

Photosynthesis catches energy and respiration releases it.

You may have learnt that plants trap energy from the sun and use it to convert low-energy carbon dioxide and water to high-energy glucose and low-energy oxygen. This process is called photosynthesis. Plants store their glucose as starch, a complex carbohydrate molecule. Animals obtain their food by consuming plants or other animals, so they are called consumers. Inside the digestive system of animals, the complex starch is broken down again into a simpler form, glucose. The glucose travels to the cells and is further broken down in the process of respiration. During respiration energy is released and made available to the cells. The cells use the energy for many chemical reactions, as well as for heat.

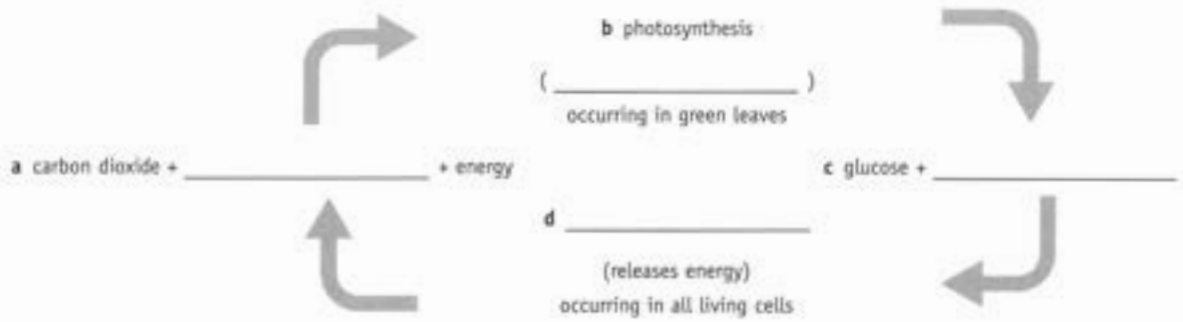
The sun's energy, which is trapped by the plants during photosynthesis, is released during respiration. In other words, respiration is an energy-releasing process. It occurs in all living things, in their cells. During respiration oxygen is used up as the glucose is broken down in many controlled reactions. Carbon dioxide and water are produced as the waste products. These become the raw materials needed for photosynthesis.

There is a cycle involving photosynthesis and respiration. In this way we do not run out of food, oxygen and carbon dioxide.

1 Write the word equation for respiration.

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2 Complete the diagram by filling in the blanks using the following labels: absorbs energy, oxygen, respiration, water.



3 Complete the summary table about respiration and photosynthesis.

Question	Photosynthesis	Respiration
What does the term mean?		
Which organisms perform the process?		
Is energy absorbed or released?		
What are the starting materials?		
What are the end products?		
Why is the process important?		

4 Write a paragraph about each process using the information from your table and from the text above.

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Respiration

15b

5 A group of students designed an experiment to demonstrate respiration. Read their recount about their experiment.

Our snail experiment – a recount

Our teacher provided us with the following items: a large 500 mL beaker, a large test tube, 3 small snails, some fine wire mesh, 2 small heavy cubes about 2 cm long, and 250 mL of limewater.

To set up our experiment we placed the three snails into the test tube and secured them with the mesh, giving them a little room to move but not allowing them to fall out when we inverted the test tube into the limewater. The open end of the test tube was balanced carefully on the blocks, making sure the liquid could come into the test tube. Initially a small amount of water, about 5 mm, entered the test tube. We marked the test tube using a permanent marker. We let the experiment stand for a day. The following day we noticed the water level in the test tube had risen by 2 cm and the limewater, especially in the test tube, had turned milky. We were all glad to see that the snails were still alive. We dismantled the experiment and let them go.

a Draw a labelled scientific diagram of the experiment.

b Explain why the limewater turned milky.

c Explain why the limewater moved up in the test tube.

d Change the recount into an experimental record. (You do not need to draw the diagram again.)

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